SOLPS-ITER v.3.0.8 Release Notes

as of 01 February 2022

These Release Notes give a summary list of the changes between SOLPS-ITER versions 3.0.7 and 3.0.8 and give information about how to transition an older SOLPS-ITER run to be continued with the 3.0.8 code version. Users are urged to read this document carefully and thoroughly before proceeding with use of SOLPS-ITER 3.0.8, and in particular follow the instructions from Section 6. Users should also consult the updated version of the SOLPS-ITER manual that accompanies this distribution and other documents posted to the SOLPS-ITER web site and Confluence page. In particular, the minutes from the SOLPS-ITER User Forums will provide more detailed information about the features discussed below. For full details of all updates, please consult the GIT repository logs. In addition to the items listed below, the new code version also contains lots of bug fixes, new documentation, and output clarification/prettifying improvements, as reported/requested by users and developers since the release of the 3.0.7 version. Some of the changes that have occurred on the Eirene side are documented in the updated Eirene manual, which can be found at www.eirene.de.

If you are transitioning from SOLPS-ITER v.3.0.6 or earlier to v.3.0.8, then you should also read and consult the v.3.0.6 and v.3.0.7 Release Notes.

The code is freely distributed upon request to civilian state and academic institutions within the ITER Member States, and to other entities when they have signed a research collaborative agreement with the ITER Organization for that purpose. The code is strictly licensed for non-commercial purposes only. Usage of the code in Switzerland or the United Kingdom and its dependencies is subject to the country's membership within EURATOM, which in both cases is currently under negotiation.

Users that have not done so already are encouraged to subscribe to the SOLPS-ITER Slack channel. Simply send an email request to the IO Responsible Officer to receive an invitation to join.

1. **How to obtain SOLPS-ITER v.3.0.8:**

   In order to update your copy of the SOLPS-ITER code suite from versions 3.0.6 or 3.0.7 to 3.0.8, use the *solps_iter_update* or *solps-iter_update_3.0.8_release* scripts. If coming from version 3.0.5 or older, run the *switch_to_master* script first. As of version 3.0.6, the SOLPS-ITER distribution does, by default, point to the *master* branch of the SOLPS-ITER GIT repository and its submodules. If you wish to point instead to the *develop* branch, then use the *solps-iter_update_develop* script. Note that the *safe_develop* branch of **Eirene** will not continue in version 3.0.8, and is going back to the *develop* naming convention. As usual, before making any such update, please make sure that any local changes you might have are committed to local GIT branches or stashed, so as to avoid them being overwritten or creating conflicts.

Users that also need the SOLPS4-5 converter program **b2sxdr** should update their copy of SOLPS4.3 as well.

For subsequent updates, users should use *solps-iter_update* or *solps-iter_update_develop*, depending on which branch they are using, and only once per update.

IMPORTANT: Because the update to v.3.0.8 involves a new structuration of the compiler configuration files, new environment modules (in particular, if you want IMAS support, it is recommended to use IMAS/3.34.0, but at least version 3.28.0), and some number of added/renamed/moved/removed source code files, it is likely that the compilations launched by the update scripts will fail the first time around. If this is the case, start afresh in a new window, load the SOLPS-ITER environment as usual by means of the *setup.csh* file, and try again to compile. If it fails once more, then do a "*gmake clean*" first and compile again. Recall that you can use "*gmake clean_XXX"* to remove all objects associated with any *XXX* build, for example *"gmake clean_solps_mpi"*.

For new users, note that it is now possible to obtain credentials to the ITER GIT server by means of a token instead of the SSH key method.

2. Changes to the physics model:

Please be mindful that the changes to the default physics model described in this section will most likely cause some evolution of existing 3.0.7 solutions. The magnitude of that change will depend on the specifics of the cases at hand. The benchmarks and sample cases available as part of the code distribution have all been recomputed using the 3.0.8 physics model.

a. Grad-Zhdanov closure scheme:

A new Grad-Zhdanov closure scheme has been implemented to compute accurately the thermal and friction force terms for arbitrary plasma compositions. Its use is governed by setting the new switch `b2tqna_Zhdanov_closure` to `1`. By default, this is not turned on.

The Zhdanov closure scheme also requires the speciation of the currents, i.e. assigning all currents, from anomalous and drift terms, to the species carrying them. Previously, all ionic currents were assumed to be carried by the main hydrogenic fuel ion species only. This splitting of the currents is now applied to the default physics model, which causes a small change in the solution for multi-fluid runs. The species-specific currents are now stored in the new variables `fchvispar_a`, `fchvisper_a`, `fchvisq_a`, `fchinert_a`, and `fchanml_a`.

As part of the work for this new feature, a missing factor of Z_{eff} was discovered in the pre-multiplier to the parallel current when computing the poloidal electron energy flux. This correction will slightly affect multi-fluid runs in which currents are computed and the default thermal and friction force model from Sytova *et al.* is employed.

b. Species-specific particle balance rescaling scheme in Eirene:

It is now possible to rescale the Eirene sources for each species separately, as opposed to for each particle type only. This is done by setting the `LSPRCL` switch in block 14 of the Eirene input file (the fifth logical on the first line) to `.true.`, and is the new recommended setting for multi-fluid runs. When building a case using **Uinp**, this switch will be activated by default.

c. Recycling of ions in Eirene at N/S domain boundaries:

The model for recycling ions at the “North” and “South” boundaries of the B2.5 computational grid (i.e. in front of the main chamber wall and into the private flux region) has been corrected to no longer include any sheath acceleration. This is imposed by the code regardless of the settings provided in block 14 of the Eirene input file. This change will affect the results from all coupled runs.

d. Time-dependent mode for B2.5:

A new solution method, activated with `b2mndt_style.eq.2`, is available. It allows for a more accurate treatment of time-derivative terms, and, as such, is called the “time-dependent” mode, even though this is somewhat of a misnomer. It changes the order of solution of the equations and imposes use of the SOLPS5.2 physics model. This feature is meant to be used without any under-relaxation, i.e. with `b2mndt_rxf` set to `1.0` (default for the time-dependent mode, otherwise the default value is `0.5`), although this mode of operation seems to be unstable for “small” machines (i.e. JET size or smaller). In that case, it is still possible to use a combination of `b2mndt_rxf` and the time-multipliers `DTXX` from *b2.numerics.parameters*, such that their product is still 1.0, and obtain good results. This feature should however still be considered to be experimental and is currently being investigated to understand the nature of the run crashes at small size.

When all the `DTXX` time-multipliers have the same non-unity value, this means that the code’s effective time-step is now `dtim×DTXX`. Therefore, in that case, the time variable will be incremented in steps of that effective value, and this will also be reflected in the result analysis scripts.

e. <u>Terms due to stochastization of the magnetic field:</u>

The model that considers a stochastic magnetic field layer has been refined and now also provides for associated radial electron heat diffusivity and electric conductivity coefficients.

f. <u>Neutral-neutral collisions and time-dependency in</u> <u>Eirene</u>**:**

The non-linear BGK treatment in **Eirene** has been wholly revisited and its numerics and accuracy improved. Similarly, the treatment of the time-dependent "census" array has been improved. These changes will affect all coupled runs where neutral-neutral collisions have been included in the Eirene neutral model.

g. <u>Ion bundling scheme in</u> <u>Eirene</u>**:**

Following the treatment of ion charge state bundling in EDGE2D-Eirene, a similar capability was added to the **Eirene** version in SOLPS-ITER. The average ion charge data is added as field `ZIB` in the *fort.31* file, and its presence is indicated by a non-zero value for `INDPRO(11)` in block 6 of the Eirene input file. For compatibility, this field is always provided, with appropriate default values, even if no charge bundling is assumed. Conversion rules for the new fields have also been added in **b2yt**.

h. <u>Radial profiles for flux limiters:</u>

It is now possible to apply a radial variation to the flux limiting coefficients `CFLMI`, `CFLME`, and `CFLMV`. For all three variables, values for the deep core and far SOL can be provided as `CFLXX_CORE` and `CFLXX_SOL`, respectively, with a *tanh* profile centred at `CFLXX_TANH_A`, and of width `CFLXX_TANH_B`, measured in metres along the outer midplane away from the separatrix. These variables can be set in the *b2.transport.parameters* file.

i. <u>Density feedback schemes:</u>

When rescaling densities using a density feedback scheme, the fluxes associated with this density are also rescaled. A more complete set of sanity checks is done at the beginning of the run to ensure the feedback scheme is self-consistent and not working at cross-purposes with other code features.

When using `NA_FEEDBACK_CHOICE` settings `1` or `4`, i.e. asking for a feedback on the total density or total particle content of a given isonuclear sequence, the densities of the Eirene species associated with that sequence are now included in the computation of the feedback quantity.

j. Prompt redeposition:

A new prompt redeposition model, based on an analysis of W data from JET, has been implemented as a new option, available by setting `b2stbr_redep_alpha` to a negative value. See the code documentation for full details. This model is however only available for **B2.5** standalone runs.

3. Changes to the input files:

a. Default reaction list:

By default, for cases involving hydrogenic and/or helium species, elastic collisions between H atoms, H_2 molecules, and He atoms (and their isotopes) are automatically included in the Eirene reaction list provided by **Uinp**. The data for these reactions is stored in the *AMMONX_Arrh-elast.tex* file which is included in the SOLPS-ITER distribution along with the other Eirene database files.

For Helium, we now recommend using the ADAS ionization rate in the Eirene input file, for both the particle and associated energy loss rates. We also now include the resonant CX reactions between He and its ions as part of the default model.

A new *Xe-el.amd* AMDS file was added that contains the `AMJUEL 0.8T` elastic collision rate between Xe atoms and hydrogenic ions.

b. B2.5 namelist files:

The `LCBS` array indicates the numbering of the core boundaries in the listing provided in *b2.boundary.parameters*. For SN and DN topologies, this array is filled by default by the code to be internally consistent. It can (and should) be modified when wanting to establish a core-like boundary in a linear case, as for instance in the `test_slab_ortho_standalone` example case.

For each namelist file that can change as a function of time, i.e. with the `<namelist>_TIME_MOD` and `<namelist>_TIME_SWITCH` variables, it is now possible to provide for a non-periodic variation, i.e. leave `TIME_MOD` unspecified. Then the namelist files will be read in sequence, moving to the next one (indicated by the value of `<namelist>_FILENAME`), when the plasma time reaches the `TIME_SWITCH` value.

The species for which one may set the density boundary condition `BCCON` to type `27` is no longer required to be the highest ionization state of its isonuclear sequence, but rather should be the species corresponding to the ionization state most likely to be dominant at the temperature expected at the core boundary.

c. Eirene **input files:**

A safety was added for the code to be able to recover in case the Eirene input file expects a non-empty *fort.13* file and it is not found. Also, if the input requires a *fort.15* file to be written out, this will be done even if the census stratum data it contains is empty. The code now also contains safeties to fetch the database files from their standard distribution location if they are found missing in the run directory.

The `NTIME0` and `NITER0` variables from block 1 of the Eirene input file are now obsolete and no longer used.

The code has been modified so that it can handle the Eirene recycling strata (in block 14) being defined as multiple segments, and the target data properly summed over all segments belonging to the same target when producing output.

The `INDSRC` switch should be set to `1` instead of `-1` for gas puff strata to better reflect the fact that the strength of the gas puff is imposed by **B2.5** by means of the `USERFLUXPARM` variable in *b2.neutrals.parameters*, and not the one provided in the Eirene input file. Conversion of old input files using **b2yt** will perform this correction automatically.

When using ADAS reaction rates, it is important to ensure that the ionization potential of the relevant species is provided as the `DPP` parameter in the reaction card. The **b2yt** converter will detect if it is missing and add it as needed, as part of the conversion of the Eirene input file.

d. **Empty** Eirene **strata:**

The recommended way to turn off an Eirene stratum is to set the number of particle trajectories it is to follow to zero in the Eirene input file. **Eirene** understands this to mean that the strata should be skipped and no data computed. Setting the flux associated with a stratum to zero may not work because these, in coupled runs, are overwritten by the fluxes provided by **B2.5**. When continuing a run where a previously active stratum has been turned off, the data associated with that stratum in the *fort.44* file is zeroed out.

Conversely, if **B2.5** specifies that a stratum has zero strength, it will transmit that information to **Eirene** and no trajectories will be followed for that stratum.

e. **Variable default value changes:**

The default value for all flux limiters has been changed from 0.0 to 1.0, meaning no modification to the fluxes.

When building a case using **Uinp**, the chemical sputtering yield specified for the Eirene input file will be copied over into the `chemical_sputter_yield` array found in *b2.neutrals.parameters*.

A new maximum default number of CX reactions has been defined in the *DIMENSIONS.F* file, given by `DEF_NRCX`. The default value is the product of the number of atomic and plasma charged species. The maximum size for equilibrium grids to be managed by **DivGeo** has been increased to 1775×1775.

The `b2stbc_coreregno` switch, indicating which segment of the South boundary in *b2ah.dat* is the core boundary, now has a topology-dependent default value, not just corresponding to the SN case.

The default value of the `DBG_EIR_MC` switch in *b2.neutrals.parameters* has been changed from `100` to `0` to better match the documentation.

f. <u>Case build-up workflow:</u>

When building an Eirene input file, **Uinp** will check whether the TRIM data files required for the various projectile-target pairs are available before adding them to block 6. Recall that, if a TRIM data file is missing for a rate **Eirene** needs, data from the default dataset in the *trim.dat* file with the nearest projectile-target mass ratio will be extrapolated instead.

When choosing an AMDS reference file in your DivGeo model, it is now possible to ask for the *Uinp_defaults.amd* file, which will provide the same list of reactions as **Uinp** would give by default if no AMDS file was given. Since the AMDS references are additive, this allows to complement the Uinp default reaction list with additional reactions from other AMDS files without losing functionality.

A new **DivGeo** topology file, called *SN-TEXT*, is available, for single-null cases in which the X-point lies on or near the midplane on the high-field side of the plasma.

The *lns* script has been modified to avoid building broken links to the equilibrium file used by **DivGeo** if it can be found in the local directory as opposed to the location indicated in the *<casename>.dg* file.

If choosing to use a bundling scheme, or a non-default set of ADAS files, the location of these files, provided at the top of the *ratadas.filelist* file, will be carried over to the `ADAS` file handle in block 1 of the Eirene input file. The **Eirene** code has been modified to look at that handle location first (if the ADAS files are not in the run directory), and then at the default SOLPS-ITER distribution location, namely *$SOLPSTOP/modules/adas/adf11*. The path given at the top of the *ratadas.filelist* file can contain the string `SOLPSTOP`, which the code will translate as the `$SOLPSTOP` environment variable. The code will also, if necessary, prepend *$SOLPSTOP/modules/*

to that path. In addition, when converting the Eirene input file with **b2yt**, the ADAS file handle will now be adjusted to reflect the value from *ratadas.filelist*.

For species Argon or heavier, the initial density for the flat profile from **b2ai** has been lowered from 10^{16} to 10^{8} m^{-3}. **Uinp** will also now automatically create the *SOLPS_FLUIDS* and *NREG* files used by the post-processing scripts.

It is now possible to use **DivGeo**, **Uinp**, and *triang* to build up a linear case. The procedure is similar to that used when building a limiter case. Consult the inline **DivGeo** documentation for details. The linear case can have either a target at both ends, or at one end or the other. However, both ends should be defined as "targets" in **DivGeo**, and later modified according to the desired boundary conditions in the Eirene input file, *b2.neutrals.parameters*, and *b2.boundary.parameters*.

The *triang* script will now check whether the **Uinp** step has completed successfully before allowing the user to proceed further by default.

When importing an equilibrium from IMAS to **DivGeo** by means of the **ids2dg** program, it is now possible to specify the occurrence of the *equilibrium* IDS one wishes to use, by means of the optional `--occurrence` (or `-o`) argument. If the occurrence number provided does not exist, the code will revert to fetching the default occurrence of the *equilibrium* IDS. Recall that the occurrence numbering starts at `0`! When looking for the input *equilibrium* and *wall* IDSs, **ids2dg** will search in the public scenario and machine description databases respectively if they are not found in the local user's filespace. If no *equilibrium* IDS is provided, only the *wall* IDS will be imported to **DivGeo**.

The **dg_minuspsi_dg** utility program, which takes a DivGeo equilibrium file and flips the sign of the magnetic flux function, is now part of the standard set of equilibrium translation programs compiled by default. Also, the **dg_symm_dg** program has been modified so that the optional *btor.dat* file (which provided the value of the vacuum toroidal field for older versions of **DivGeo**) need not be specified when also wanting to give, by means of the fourth optional (numerical) argument, the amount of asymmetry to apply to the equilibrium.

The default location of the DivGeo input files can now be modified by means of the `DGSONNETMASK` environment variable.

The **Carre** grid generation program will now check that the requested grid sizes do not exceed the maximum sizes provided in the *DIMENSIONS.F* file, so as to avoid later problems when attempting to launch a run with over-sized grids. If running in stand-alone mode, the maximum **Carre** grid size allowed is 500 poloidal points by 200 radial points.

When looking for a Carre or Sonnet grid file to convert into a *b2fgmtry* file, the **b2ag** program will now also look in the default directory used for central installations, namely *$SOLPSTOP/modules/Carre/meshes/$DEVICE/$USER* and *$SOLPSTOP/modules/DivGeo/device/$DEVICE/$USER*, respectively.

4. New features:

a. New configuration files and submission scripts:

Configuration files for compiling the code and submitting runs on the SPBSTU cluster in St. Petersburg and the CUMULUS cluster at CCFE are now available. The corresponding submission scripts are *spbstusubmit* and *cumulussubmit*. Use of the NetCDF library at JET was updated. New MARCONI *gfortran* and ORNL *gfortran* configuration files have also been provided.

The ITER configuration files and submission scripts have been modified to be able to function on both the older HPC cluster CentOS7 nodes and the newer SDCC cluster with CentOS8. Only the latter is fully supported, HPC usage being limited to older applications with backward compatibility issues.

New configuration files and submission scripts have also been created for the JFRS computer environment in Japan, using Intel *ifort64*, *gfortran*, and Cray compilers.

It is now possible to specify the number of nodes over which the run is being submitted by means of the `-N <number_of_nodes>` option, valid for most of the submission scripts.

b. New default compiler behaviour:

For the most common compilers (Intel *ifort64*, *gfortran*, and *pgf90*), the shared switches for all environments have been assembled into a *config.common.*`${COMPILER}` file, which the *Makefile*s consult in addition to the *config.*`${HOSTNAME}.${COMPILER}` files. The common files are read before the particulars for each local host environment.

If the graphical libraries (NCAR, GR, and GKS) are not present, the *Makefile*s will detect this, print out a warning message, and not attempt to compile the executables requiring graphics.

Compilation when coupling to **IMPGYRO** (using *gmake solps_ig*) automatically implies MPI compilation as **IMPGYRO** relies on MPI parallelization.

Linking to the MSCL library has also been modified so that it only occurs when actually necessary. The recommended MSCL version is number 1.2.2, in which various issues related to type mismatches (due to the usage of antiquated FORTRAN features) have been resolved and corrected.

The *Makefile*s will also check which version of the compiler is being used and adapt the compilation flags accordingly. If the compiler has no MPI option, or the Motif library is not available, the dependencies and object list files for the corresponding objects will not be built. The *Makefile*s will also check whether the MPI system variables are to be included as a module or as an include file depending on the local MPI version available and use appropriate linking instructions.

The *Makefile*s will also extract from the environment the version numbers for the IMAS, UAL, and GGD modules. These will then be passed as pre-processor defines to the code before compilation.

c. New `SOLPSTOP_FORCE` environment variable:

A new `SOLPSTOP_FORCE` environment variable can be defined, to override the settings of the local *SOLPSTOP* file in or above the run directory, and force the use of code executables from a different SOLPS-ITER tree. To know which `SOLPSTOP` value the submission scripts and run *Makefile*s will use, one can run the *determine_SOLPSTOP* script.

d. Standardization of mathematical and physics constants:

A new `Fundamental-Constants` module has been built at ITER, and made publicly available to all, that provides many useful mathematical and physical constants as a set of standardized include files. These allow for usage of CODATA values from the 2002 to 2018 assessments. Since **B2.5** was already using the CODATA 2018 values, this does not lead to any change in the solution, but the *Makefile* checks at compilation time whether the module is loaded, and, if it is, relies on its contents. If not, but general IMAS modules are loaded, then the code relies on the `imas-constants` module.

e. Changes to post-processing scripts:

The *2d_profiles* and *2d_profiles_extended* scripts normally default to averaging over the last 10 time-points stored in the *b2time.nc* file. The scripts now check if *b2time.nc* does in fact contain the requested number of points, and if not, use all the time-points available, instead of returning an error.

The *respo* script has also been modified to be able to work with both potential equation solution methods, using either the single-pass (`b2tfhe_vis_per.eq.1.0`, default, calling *b2npp7*) or iterative (calling *b2nppo*) solution methods.

A new **nc_reduce** utility program allows to reduce the size of very large NetCDF files by only sampling the time-dependent data at regular intervals or just picking up a certain number of time points from the end of the file. See the Manual for full details.

The MDSPLUS database output format version number has been incremented to `29`, to reflect a correction in the radiation fields `emolrad` and `eionrad`, for which the total is now being written out, as opposed to only the contribution from the first stratum.

When converting a *b2mn.dat* file using **b2yt**, the comments following the switch definitions and on separate lines will now be kept.

f. Changes to conversion programs:

When converting a case from **SOLPS4.3**, a new setting of `b2mndt_style.eq.-1` is provided, that allows for writing a *b2fplasma* file and IDS output containing the equivalent data to the old *B2SXDR* file, without a recomputation of the fluxes, which sometimes led to spurious values at the targets. Conversion of an old SOLPS4.3 ITER result will automatically create a dedicated IMAS data-entry. This is the new recommended setting.

g. Continuous integration (CI) testing:

New commits to the code now get tested against a series of standardized reference cases. These include **B2.5** stand-alone cases with and without drifts, a Helium plasma case, coupled cases with and without MPI parallelization, and a DN topology case. This latter DN case is a new example, named *DIIID_3106_DN_D+C*, and is available as part of the standard set of distributed examples. It is a DIII-D-like geometry, with a disconnected DN topology and active X-point at the bottom, which uses D+C species, including chemical sputtering.

5. Changes to the output:

a. IMAS support:

When writing out IMAS data-entries, Data Dictionary versions up to 3.34.0, UAL up to 4.9.3, Generalized Grid Description up to 1.10.0, and AMNS up to version 1.3.5 are supported. New features include:

- The `code` and `library` information for each IDS;
- The projected edge areas, cell volumes, and connection lengths associated with GGD objects in their respective `geometry(:)` nodes as well as the corresponding `geometry_content` identifiers;
- The `midplane_id` flag to define the precise location of the midplane;
- The *divertors* IDS;
- Removal of the obsolete *numerics* IDS;
- Renaming of the `local%divertor_plate` node from the *summary* IDS as `local%divertor_target` (starting with DD/3.35.0);
- The `j_total` and `j_parallel` nodes in the *edge_profiles* IDS;

- The neutral velocities in the *edge_profiles* IDS (provided the `P[UV][XY]` arrays are available in the *fort.46* file);
- The `temperature_reference` node in the *wall* IDS;
- The `flux_expansion`, `power_loss`, and `radiated_power` nodes in the *summary* IDS;
- A new "private" grid subset, with index `−101`, to contain the cells used for the neutral pressure calculation, if the `LPFR[BT]_[IO]` variables are defined in *b2.user.parameters*;
- The plasma current, q_{95}, and vacuum toroidal field values are imported from the *equilibrium* IDS, if available, to be used in all other edge-related IDSs, possibly rescaled by switches from *b2ag.dat* if appropriate;
- Support for limiter configurations.

For fluxes given in units of #.m^{-2}, the area implied by the division depends on context. For quantities given in the *divertors* IDS, the relevant area is the area of contact between the plasma and the divertor target surface. However, for quantities stored on GGD objects as in the *edge_profiles* and *edge_transport* IDSs, the relevant areas are those normal to the local magnetically-aligned coordinate system, so that the physical quantities remain independent of the grid details. The used areas are found in the `geometry(:)` node associated with each GGD object.

Note that the database for ITER uses the uppercase name, so the code will internally change the name deduced from the `$DEVICE` environment variable `iter` to `ITER`. When looking for an old file, however, it will scan the second location if the first one does not yield a result, but new files will always be written to the `ITER` database location.

It is possible, during a run, to output IDSs at regular intervals, measured in terms of plasma time and/or number of code iterations, or to enforce an IDS write-out at the end of run, by means of the `b2mndr_ids_save` and/or `b2mndr_ids_time` switches. If the run is a time-continuation of an existing run, the IDSs will be appended as new time-slices to the existing IMAS data-entry. It is also possible to output in IDS format the batch-averaged solution (obtained from the Eirene averaging speed-up scheme), by means of the `b2mndr_ids_av` switch. The batch-averaged plasma solution and associated averaged Eirene sources are written as occurrence `1` (as opposed to the default occurrence number `0`) of the `edge_profiles` and `edge_sources` IDSs. If only the batch-averaged data is written out, the associated `summary` IDS will contain the averaged plasma parameters at the outer midplane separatrix and divertor strike-point locations, instead of the data from the last snapshot.

Recall that the `ion(:)` data structure in IMAS is meant to represent entire isonuclear sequences, while data related to each individual ionization state is stored at the `ion(:)%state(:)` level.

A new **b2_ual_rewrite** utility program allows for rewriting an existing IMAS data-entry, using data from the same original run, but allowing for updates in Data Dictionary version, any eventual new quantities, or bug fixes. It is possible to either overwrite the same IMAS data entry or to write a new one with a different run number. In this latter case, the utility script *IDS_yaml_replace* can be used to create a new *yaml* file and modify the old *yaml* file so that the old data entry is now marked as obsolete and replaced by the new one, and that the new one is marked as replacing the old one.

b. Debugging output from kinetic neutrals:

The output related to the additional tallies from block 10A of the Eirene input files, computed in the *eirene_uptusr* routine, has been corrected.

The `DEBUG_FLAGS` array defaults from *b2.neutrals.parameters* have been set so that all output is now disabled by default, and additional documentation has been provided to explain the meaning of the various flags in the array. Conversion rules for this array have also been included in **b2yt**.

c. Eirene **plots:**

When running **Eirene** in stand-alone mode and with multiple time-steps, the trajectory plot, if requested, will now be produced once per time-step, each time on a separate frame. The aspect ratio for these trajectory plots has also been adjusted so that it always has a minimal practical height or width such that the labels are not cropped away.

Also, for **Eirene** stand-alone runs using the `LEVGEO=1` (Cartesian grid) or `LEVGEO=2` (nested ellipses) options, the geometry (and trajectory, if requested) plots will now include both sides of the symmetry axis, i.e. also show the region with negative *R* values.

d. New *fort.44* quantities:

A new array `ewldt_res` has been added to the *fort.44* file, and the `jvft44` format identifier increased to `20201006`. This array contains the total heat load from Eirene particles onto the B2.5 grid edge surfaces and is used for some post-processing **b2plot** quantities. Conversion rules for this array have been added to **b2yt.**

The power load due to neutrals now includes a term due to the recombination into molecules of the thermally recycling ions, when applicable. This change also applies to the SOLPS4.3 code.

e. New b2plot options:

It is now possible to label the axes of **b2plot** graphs using the *xlab* and *ylab* functions.

The Coulomb logarithm functionality *lnlam* can now take an optional integer argument. This argument allows to choose between the Braginskii formula (default), or one of the Wesson formulae for ion-ion, ion-electron, or electron-electron collisions. The Coulomb logarithm formula (instead of a constant) will also now be used when computing the Chang-Hinton χ_i when invoking the *chhichii* command.

Non-standard isotopes of heavy species now have their labels decorated with a leading superscript giving the isotope mass when shown in graph labels, such as ^{3}He.

The *trgl* option is obsolete and has been removed.

f. New *b2wdat_iout* output options:

If setting `b2wdat_iout` to `2` or `3`, the user will now obtain a set of output files (in the *output/* directory below the run directory) that will contain upstream and target profiles useful for detachment physics studies and a listing of the primary code quantities, respectively. These files can be appended at regular intervals of B2.5 iterations by means of the `b2wdat_append` switch.

g. Changes to tracing files:

The particle balance diagnostics in the *blnn_SPb.trc* file have been modified to include the particle sources due to CX occurring with Eirene species.

h. Changes to averaged solution files *b2faver[ie]*:

The averaged solution files *b2faver[ie]* now only contain, when considering the source arrays `s[ch|he|hi|mo|na|ne]_mean` and `e_s[ch|he|hi|mo|na|ne]`, the average and variance values for the total sources, respectively, not the average and variance values for each source component. The `sne_mean`, `sch_mean`, `e_sne`, and `e_sch` fields have been added to the file. Conversion rules in **b2yt** will create a file with the new format, and other relevant code routines can read the older format and convert it on the fly, but **b2yt** conversion to the new format may not be correct for cases using charge state bundling.

6. Getting a case restarted with v. 3.0.8:

As a consequence of all the changes above, in order for users to take full advantage of the new 3.0.8 features, users wishing to continue their older runs should keep in mind the issues listed below. If running an existing 3.0.7 case, continuation of this case with the 3.0.8 version should be possible without any special additional preparations, although it is recommended to correct the input files following the information given in Section 3 of this document or run a **b2yt** conversion step to apply all automated corrections. Of course, when converting older cases or building new cases from scratch, the **Uinp**, **b2yt** and **b2sxdr** programs have been modified to produce input files already compatible with version 3.0.8, which should not preclude a visual inspection of all relevant input files, as is best practice.

1. If you are restarting a case that was computed using code from before November 2019, you should delete the existing *fort.1[0-5]* files as there is an internal change in **Eirene** that has modified the content of these files.
2. If you were previously using NetCDF3, you may not be able to append to your older **.nc* files and would then need to delete/rename them.

7. Additional notes for developers:

i. A large amount of effort has been devoted to bring the source code more in accordance with the "good housekeeping" rules as posted here. The driving philosophy has been to minimize the amount of compiler warnings to obtain as uncluttered as possible a compilation process and to maintain portability. In particular, the use of obsolescent features such as algorithmic or computed `GOTO`s, `H` format descriptors, alternate returns, statement functions, subroutine entries, etc… has been replaced with case constructs, contained functions, additional calling arguments, modularization of routines, etc… Many occurrences of unused arguments to subroutine calls have also been removed. Most unused labels have been removed. Attention has been paid to avoid any type mismatches in code operation, as modern compilers are becoming increasing intolerant of them.

ii. Many commonly used geometry-related statement functions, such as `cr`, `cz`, `abs_diff`, `dist`, `distc`, `disx`, `disy`, `hy1`, `pit`, `points_dist`, `sx`, `sy`, etc… have been moved to *b2mod_geo.F* or *b2mod_geo2.F* and are now full-fledged stand-alone functions. The `init_region` and `init_connectivity` routines are now stored in those modules. New functions `divide_by_contact_areas` and `divide_by_poloidal_areas` are also provided in *b2mod_geo.F*. The new function `get_connection_length` has been added to *b2mod_interp.F*.

iii. Similarly, the new *b2mod_math.F* module now contains most of the statement functions of a mathematical nature. These include `addup`, `cosang`, `expu`, `harm`, `hybr`, `ltst`, `machsfr`, `med`, `norm`, `sinang`, `trim1`, `trunc`, `upwind`, uxcm, etc... The *b2mod_math.F* module also initializes the machine-dependent smallest non-zero real cutlo and its logarithm cutll, used to avoid denormalized values in the calculation, particularly when running in debug mode.

iv. Several plasma-related secondary quantities and related functions have been added to the *b2mod_plasma.F* module. These include the mass density roxa, the electron velocity variable ue, functions summing fluxes along boundaries and mapping velocities to cell faces, and the routines used to initialise, increment, and decrement the drift term multipliers.

v. Quantities related to plasma species (amtol and the lnext function) have been moved to the *b2mod_b2cmpa.F* module, and the lrtnext function to the *b2mod_b2cmrc.F* module. Transport functions formulae have been moved from *b2tqce.F* to *b2mod_transport_fun.F*.

vi. The matrix solver routines *il[lu]tern_[7|9|11]_solps.F* have been modularized into modules *b2mod_il[lu]tern[_7|9|11].F*.

vii. The calling stack management routines subini and subend have been moved to the *b2mod_subsys.F* module.

viii. Among other routines that were modularized, the conversions include *b2agsp.F* to *b2mod_aggs.F*, *b2bln[cems].F* to *b2mod_b2bln[cems].F*, *b2file.F* to *b2mod_file.F*, *b2trace.F* to *b2mod_trace.F*, *b2usrtrc.F* to *b2mod_usrtrc.F*, *b2wr[int|sep|srt].F* to *b2mod_wr[int|sep|srt].F*, *batch_av_all.F* to *b2mod_batch.F*, *eirene_mc_F* to *b2mod_eirene.F*, *gnmap[23].F* to *b2mod_gnmap[23].F*, *ipmain.F* to *b2mod_ipmain.F*, *lbfill.F* to *b2mod_lbfill.F*, *lwimai.F* to *b2mod_lwimai.F*, *lwmain.F* to *b2mod_lwmain.F*, *running_average.F* to *b2mod_average.F*, and *xerset.F* to *b2mod_xerset.F*.

ix. The version number reading and writing routines, cfverr and cfverw, have been moved to the *b2mod_version.F* module. This module also now contains the new variable b2faveri_version for the version number of the averaged solution files.

x. The *test_couple.F90* routine was moved from the **Eirene** to **B2.5** repositories.

xi. To allow for the usage of the FORTRAN90 include files from the Fundamental-Constants module in the **B2.5** repository, *b2mod_constants.F* has been moved from the *modules/* to the *ids/* directory and its extension changed to *F90*. In the **DivGeo** repository, the *eqdim.inc* include file has been replaced by the *eqdim.f90* module. Inside the code, the presence of the Fundamental-Constants module is linked to the definition of the CONSTANTS_PROVIDED pre-processor pragma.

xii. The *infcop.F* routine has been moved to the *eirmod_infcop.F* module and the variable definitions from the *eirmod_extrab25.F* module have been split off and moved to a new *eirmod_wneutrals.F* module, which also includes what used to be in the *wneutral_fluxes.F* routine.

xiii. In **Eirene**, the MPI broadcast of module variables has been moved from the *broadcast.F* routine to individual routines in each affected module, named EIRENE_BROADCAST_<module_name>.

xiv. The input profile routines `EIRENE_PROFE`, `EIRENE_PROFN`, `EIRENE_PROFR`, and `EIRENE_PROFS`, have been grouped in a new module file *eirmod_profiles.f*.

xv. Many new routines have been introduced as part of the Grad-Zhdanov closure scheme. These include *b2mod_b2mrel.F*, *b2mod_b2zhco.F*, *b2mod_zhtrtf.F*, *b2mod_zzh.F*, *b2nxfv_zh.F*, *b2siav_zh.F*, *b2xz[11|12|22].F*, *b2xzf[12].F*, *b2xzmc.F*, *b2trc[a|c|n|s|v].F*, *b2trfl_[an|vis].F*, *b2trfr.F*, *b2trhrtf_an.F*, *b2trht.F*, *b2trht_an.F*, *b2trhw.F*, *b2trnu.F*, *b2trtf.F*, *b2trvh.F*, *b2trvis_an.F*, and *b2trzh.F*.

xvi. New routines for the computation of transport coefficients associated with magnetic field stochastization can be found in *b2trsc.F* and *b2trsh.F*.

xvii. The **B2.5** “time-dependent” mode introduces the new routines *b2news_m.F* and *b2s[c|h|m]dt.F*.

xviii. A compiler pre-processor definition is used to rename the *samax* routine to *damax*, to avoid a name clash with SOLPS4.3 when converting older run files.

xix. The interface to several MDSPLUS utility routines has been modified to avoid compiler errors about mismatched argument types when compiling with GCC/10.2.0.

xx. It was discovered that the NVIDIA PGF90 compiler handles formatted reads of a sequence of integers, when there are less numbers in the string being read than the read statement demands, differently. Unlike the Intel and GNU compilers, it does not assign zeroes to the extraneous integers but rather moves to the next line in the input file in an attempt to find them there. This can break the formatted read of the Eirene input file. Therefore, a new utility function `fix_integer_input` has been created (and added to the *input.F* routine) that can be used to prevent such errors when it is expected that some integers being read from the input may not be present.